

Refresh and Content Opportunity Scoring Using Historical Search Signals: A Leakage-Safe Grouped-Client Machine Learning Study

Zafar Ullah

Department of Computer Science, University of Engineering & Technology Mardan, Pakistan

Corresponding author: zafarullah1385@gmail.com

ORCID: Not available

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Abstract

Content teams frequently face more pages than they can investigate in a single review cycle, creating a practical need for transparent prioritisation rather than automatic editing. This study evaluates whether historical search-performance signals can rank eligible content pages for human refresh and content-opportunity review. The analysis uses 9,841,378 anonymised daily records from the March 2026 FlyRank ML Internship warehouse partition, aggregates them to page level, separates a March 1-15 feature window from a March 16-31 outcome window, and retains 34,038 eligible client-page records across 32 anonymised clients. A frozen operational baseline, Logistic Regression, and Random Forest are compared on the same grouped-client holdout with zero client overlap; leakage is additionally tested by deliberately introducing a future-derived field. Logistic Regression measured Precision@50 of 0.640 compared with 0.520 for the baseline, while the validation positive base rate was 0.3008 and Average Precision was 0.3939. The top-50 model queue contained 32 pages meeting the later decline proxy. Permutation importance indicated that CTR, position band, average position, and position volatility were the strongest measured sources of model reliance. The findings are observational and directional: they support human decision-support for review prioritisation but do not establish causal refresh effects, guaranteed recovery, or suitability for automatic content changes.

Keywords: search engine optimization; content refresh; content opportunity scoring; grouped validation; data leakage; logistic regression; decision support

Paper type: Technical Report

Data and code availability: The public repository contains executed notebooks, aggregate metric receipts, figures, and the deployed research page. Raw warehouse data are not redistributed. Dependency constraints are declared in the repository requirements file.

1. Introduction and Problem Statement

Search engine optimisation (SEO) decisions are often made under constrained editorial capacity. A portfolio may contain thousands of pages, while a content team can inspect only a small queue during a weekly or monthly review cycle. The operational problem is therefore not merely to describe search performance, but to decide which pages should be inspected first and why. FlyRank's portfolio research similarly emphasises simple validated comparisons, protection of strong page-one assets, and refresh review before visible decline [1].

Search metrics are noisy and interdependent. Impressions describe exposure, clicks describe realised search visits, CTR is conditional on exposure and search-result position, and average position can hide volatility across days or queries. External SEO research reports statistically significant associations for some on-page and technical signals, but the measured effects are generally small and observational [2]. Single-site intervention studies can demonstrate practical improvement after on-page work, yet they do not by themselves establish that the same action will work for every page or client [7].

This study frames content maintenance as a ranked decision-support problem. It asks whether signals available before a review decision can identify a top-50 queue with a higher observed later-decline rate than a transparent hand-written baseline. The primary output is a rank and an action rationale for human inspection. It is not an automatic publishing, deletion, redirect, or rewriting system.

1.1 Research Question

Can historical prediction-time search-performance signals rank eligible pages so that the first 50 pages inspected by a human content team contain a higher observed later-decline rate than a transparent fixed-rule baseline?

1.2 Study Contributions

- A public-safe page-level analytical contract with separated feature and outcome windows.
- A same-row comparison between a frozen operational baseline, Logistic Regression, and Random Forest.
- Grouped-client validation in which no anonymised client appears in both training and validation.
- An executable leakage audit demonstrating how a future-derived field creates near-perfect but invalid performance.
- Error analysis that distinguishes ranked queue quality from weak threshold classification.
- A human-reviewed content action playbook with reason codes, no-go automation rules, and monitoring triggers.

2. Related Work

The reviewed literature falls into four relevant streams: direct SEO signal studies, content and organisational adaptability studies, learned ranking and decision-support systems, and emerging generative-search research. These streams provide useful methodological and substantive context, but they usually address different units of analysis or validation questions.

2.1 Direct SEO Signals and Content Interventions

Manetti's Google Italy study analysed 6,824 unique pages and 57 technical and on-page signals across 841 queries. Ten signals reached statistical significance, with keyword presence in titles, URL slugs, headings, and meta descriptions among the strongest; however, correlations were small, cross-sectional, and

not causal [2]. The Tourism Village case study used white-hat on-page changes and reported improved visibility over a three-month evaluation period, providing practical evidence that systematic review can matter while remaining limited to one website and one intervention context [7].

2.2 Adaptability and Content Strategy

Reji's survey of technology firms found that content strategy quality and keyword strategy were significant predictors in a multivariable model of SEO adaptability, while several other organisational constructs were not significant in the full model [4]. This supports repeatable editorial processes but uses self-reported organisation-level data rather than page-level search outcomes.

2.3 Learned Ranking and Automated SEO Systems

Alhenshiri, Jelwal, and Badesh compared a learned semantic reranking layer with the original search ranking on the same candidate results and reported improvements in time, clicks, query reformulation, and preference for 14 participants [3]. SmartSEO presents an automated NLP and predictive-analytics pipeline for keyword, readability, ranking, and intent analysis [8]. These studies support the idea that learned systems can improve prioritisation, while also illustrating the need for transparent baselines, independent validation, and careful interpretation of system-generated recommendations.

2.4 AI-Mediated Search and Governance

Large language model answer engines shift visibility from ranked lists toward retrieved evidence and generated responses. Muñoz Tebar proposes an AI Search Optimization framework centred on semantic publishing, entity authority, structured knowledge, machine-readable content architecture, and verifiability [6]. Wen et al. identify underexamined risks including concentrated influence, undisclosed commercial influence, and evaluation gaps between offline research and deployed systems [5]. These concerns strengthen the case for human review, auditability, and deployment-aligned monitoring.

2.5 Research Gap

Among the reviewed literature, limited work combines real multi-client organic-search data, separated feature and outcome windows, a page-level future-decline proxy, grouped-client validation, a transparent operational baseline, an executable leakage audit, interpretable model comparison, and a human-reviewed action playbook in one reproducible workflow. This study addresses that applied methodological gap. Its contribution is not a new learning algorithm; it is a defensible validation and decision-support design.

Table 1. *Selected literature and the gap addressed by this study*

Study	Design	Use in this paper	Remaining gap
FlyRank [1]	Portfolio-level observational comparisons	Freshness, position, and content review context	Not a page-level grouped future-decline model
Manetti [2]	Cross-sectional correlation; 6,824 pages	Feature rationale and non-causal framing	No future outcome or action queue
Alhenshiri et al. [3]	Controlled reranking experiment	Same-candidate baseline comparison	User search ranking, not content maintenance
Reji [4]	Organisation-level quantitative survey	Content strategy and adaptability	Not page-level or time-separated
Wen et al. [5]	Position and governance analysis	Monitoring, disclosure, and deployment risk	No SEO decline-prediction evaluation
Muñoz Tebar [6]	Conceptual technical report	AI-readable content and trust context	Conceptual rather than empirical validation

3. Data and Public-Safety Contract

The study uses the March 2026 partition of the anonymised FlyRank ML Internship warehouse table containing daily content performance. The source partition contains 9,841,378 daily rows, 55 anonymised source clients, and 331,437 unique client-page pairs. Aggregation produces one analytical row per anonymised client-page pair.

The design deliberately separates the prediction moment from the later evaluation window. Features use March 1-15, 2026. The proxy outcome uses March 16-31, 2026. The June sample is not used for development because the final month is treated as a natural outcome window in the internship data guidance.

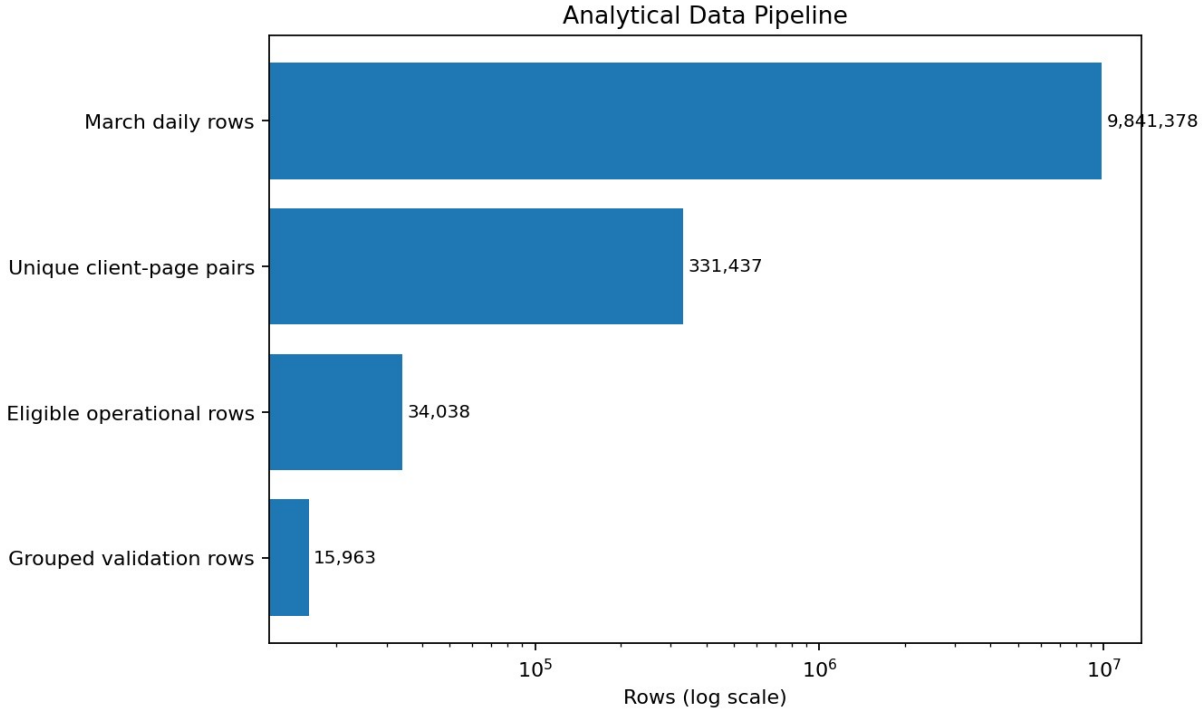


Figure 1. Analytical data pipeline from daily warehouse rows to the grouped validation set. Note. The horizontal axis uses a logarithmic scale because the pipeline reduces millions of daily rows to a page-level review population.

3.1 Eligibility Criteria

- At least 500 feature-window impressions.
- At least five feature-window days with available Search Console data.
- At least five outcome-window days with available Search Console data.
- Feature-window average position greater than 0 and no greater than 20.
- A valid feature-window CTR.
- A valid position band: Top 3, Page 1, or Page 2.

After filtering and deduplication, the operational population contains 34,038 page rows across 32 anonymised clients. This is the population to which the reported model and baseline results apply.

3.2 Public-Safety and Exclusions

No client names, raw URLs, domains, query text, private business fields, access tokens, or direct identifiers are included in public tables or figures. Pseudonymous client IDs are used only to create the grouped split. Pseudonymous content IDs are used only for deduplication and private queue regeneration. Neither identifier is a model feature.

Table 2. Data windows, grain, and exclusions

Contract item	Definition
Unit of analysis	One anonymised content page for one anonymised client
Feature window	1-15 March 2026
Outcome window	16-31 March 2026
Source rows	9,841,378 daily rows
Operational population	34,038 eligible page rows across 32 clients
Grouping key	Anonymised client identifier; grouping only
Excluded from features	Outcome fields, target, IDs, URLs, queries, product flags, and decision scores

4. Methodology

4.1 Target Proxy

The target is a retrospective decline proxy based on average daily impressions. A page is labelled positive when its outcome-window average daily impressions are less than 80% of its feature-window average daily impressions.

Decline proxy = 1 if outcome daily impressions < 0.80 x feature daily impressions; otherwise 0.

The label is useful for ranking later decline risk, but it is not equivalent to poor content quality, business loss, or a proven need to refresh. Search demand, seasonality, SERP changes, indexing conditions, and competitive changes may contribute to the observed outcome.

4.2 Prediction-Time Features

Table 3. Final leakage-safe feature set

Feature	Definition	Why available at decision time
Log feature impressions	$\log(1 + \text{total feature-window impressions})$	Visibility known before the outcome window
Feature clicks	Total feature-window clicks	Realised search visits known before the decision
Feature CTR	Clicks divided by impressions	Engagement relative to exposure
Average position	Mean feature-window Search Console position	Observed visibility level
Active days	Days with positive impressions	Stability of measured exposure
Position volatility	Daily standard deviation of position	Instability known in the feature window
Position band	Top 3, Page 1, or Page 2	Interpretable categorical context

4.3 Frozen Operational Baseline

The ML-07 baseline is preserved without tuning on the validation set. Training data define the median CTR for each position band and the empirical visibility distribution. CTR weakness receives 70% of the

score and visibility percentile receives 30%. Pages at or above the expected CTR for their band receive zero baseline score.

$$\text{Baseline score} = 100 \times (0.70 \times \text{CTR weakness} + 0.30 \times \text{visibility percentile}), \text{ with zero score when CTR weakness} = 0.$$

4.4 Models and Preprocessing

Logistic Regression is the pre-declared primary model because the target is binary, the predicted probability naturally supports ranking, and the fitted relationships remain easier to audit than a large nonlinear ensemble. Numeric fields use training-only median imputation and standardisation. Position band uses training-only most-frequent imputation and one-hot encoding.

Random Forest is included as a controlled nonlinear comparison. It is selected only if it improves Precision@50 by at least 0.05 over Logistic Regression and does not reduce Average Precision. This rule prevents complexity from being rewarded automatically. Random Forests can capture interactions and nonlinearities [11], while the implementation uses scikit-learn pipelines for reproducibility [10].

4.5 Validation Design

The retained split is a grouped client holdout with a 25% test fraction and random seed 42. Every page belonging to an anonymised client is placed entirely in training or validation. The training set contains 18,075 rows from 24 clients; the validation set contains 15,963 rows from eight clients; client overlap is zero.

A random-row split is reported only as a contrast. It places 24 clients in both training and validation. Although the random split produced higher Average Precision and ROC-AUC in this run, its Precision@50 was lower than the grouped estimate. The grouped design is retained because it directly evaluates performance on unseen client contexts rather than because it necessarily produces a lower number.

Table 4. Random-row contrast and retained grouped-client validation

Design	Val. rows	Val. clients	Overlap	Base rate	P@50	AP	ROC-AUC
Random row split	8,510	24	24	0.3089	0.500	0.4220	0.6437
Grouped client split	15,963	8	0	0.3008	0.640	0.3939	0.6175

4.6 Evaluation Metrics

Precision@50 is the primary metric because the intended user has limited review capacity. It measures the observed positive rate among the 50 highest-scoring validation pages. Precision@20, recall@50, Average Precision, ROC-AUC, and the validation base rate are also reported. Precision-recall metrics are particularly informative when the positive class is not the majority and the operational concern is the quality of the highest-ranked items [12].

4.7 Leakage Audit

The honest feature set excludes the later outcome, outcome availability, the target, trend fields, existing product scores, action flags, client IDs, and content IDs. To test the audit harness, the analysis deliberately creates `decline\_ratio`, a future-derived ratio of outcome daily impressions to feature daily impressions. This field produces near-perfect validation and is then removed.

Table 5. *Honest versus deliberately leaky validation*

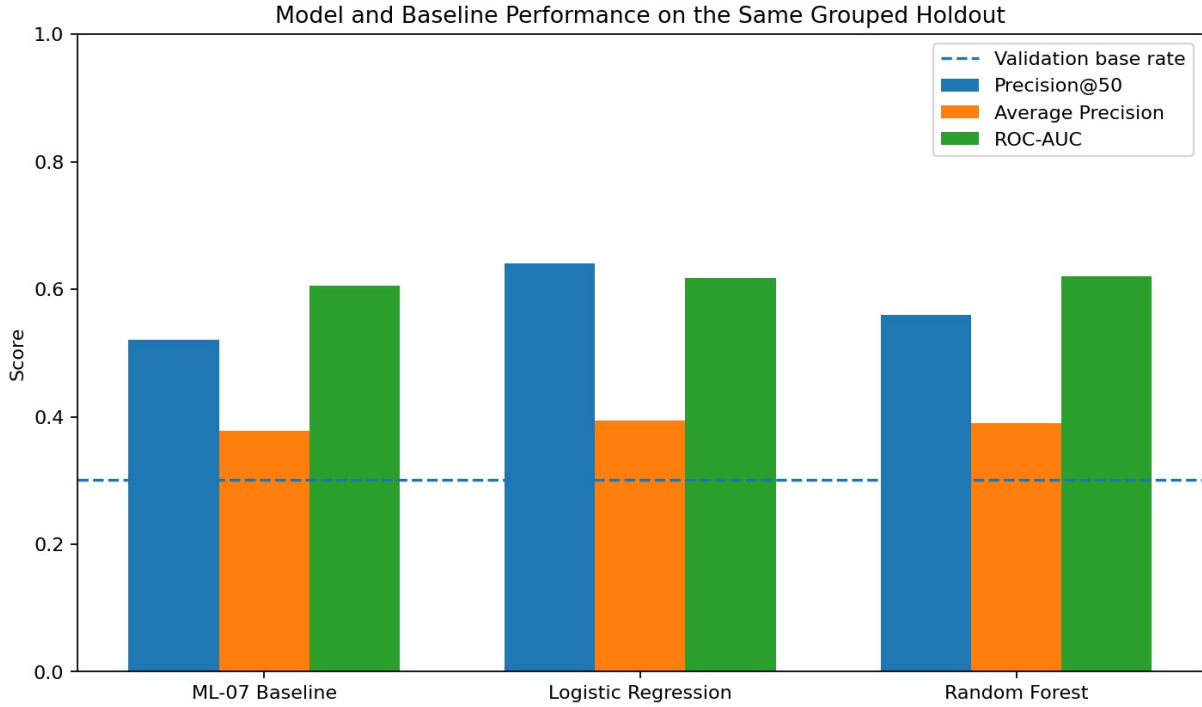
Feature set	P@50	Average Precision	ROC-AUC	Decision
Honest pre-outcome features	0.640	0.3939	0.6175	Retained
Features plus future-derived decline ratio	1.000	0.9999	1.0000	Rejected as leakage

5. Results

All three methods are evaluated on exactly the same grouped-client validation rows. The validation positive base rate is 0.3008. The frozen baseline measured Precision@50 of 0.520. Logistic Regression measured 0.640, an absolute increase of 0.120, while Random Forest measured 0.560.

Table 6. *Model and baseline results on the retained grouped-client holdout*

Method	Base rate	P@20	P@50	Lift@50	Recall@50	AP	ROC-AUC
ML-07 Baseline	0.3008	0.550	0.520	0.2192	0.0054	0.3776	0.6059
Logistic Regression	0.3008	0.500	0.640	0.3392	0.0067	0.3939	0.6175
Random Forest	0.3008	0.550	0.560	0.2592	0.0058	0.3893	0.6205

**Figure 2.** *Model and baseline performance on the same grouped-client holdout.* Note. Precision@50 is the primary operational metric; the dashed line is the validation positive base rate.

5.1 Model Selection

Random Forest achieved a slightly higher ROC-AUC than Logistic Regression (0.6205 versus 0.6175), but it did not improve Precision@50 by the pre-declared 0.05 margin and its Average Precision was lower. Logistic Regression therefore remains the selected model. This result illustrates why complexity should not be selected from one metric alone.

5.2 Top-50 Queue

The selected model's top-50 queue contained 32 pages meeting the later decline proxy and 18 pages not meeting it. This corresponds to Precision@50 of 0.640. The baseline top-50 corresponds to 26 positive pages. The difference is meaningful for a team reviewing a fixed number of pages, but it does not imply that all 32 positive pages require the same editorial treatment.

5.3 Feature Reliance

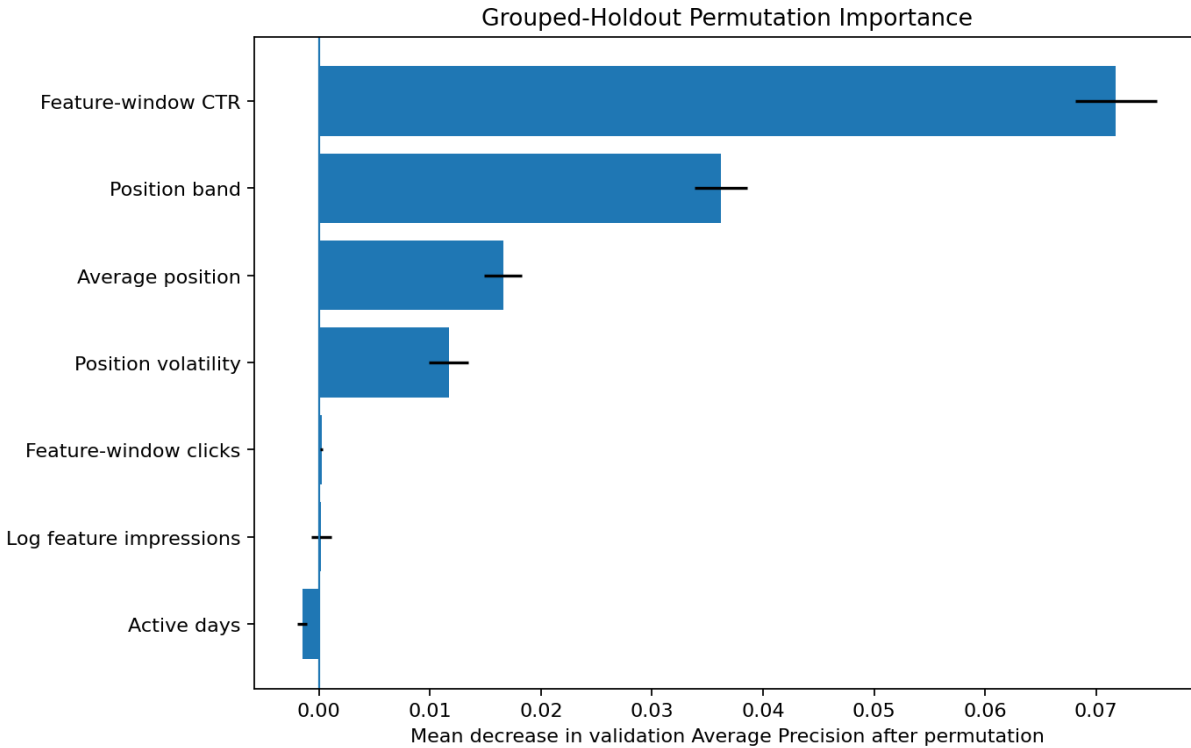


Figure 3. Grouped-holdout permutation importance for the selected Logistic Regression pipeline. Note. Importance is the mean decrease in validation Average Precision after shuffling one input. Position band is derived from average position, so their contributions are related.

CTR was the strongest measured source of model reliance, followed by position band, average position, and position volatility. Clicks and log impressions had near-zero permutation effects, and active days had a small negative mean importance. These results describe the fitted pipeline on this holdout. They do not establish that changing CTR or position causes a later change in impressions.

6. Error Analysis and Interpretation

A probability threshold of 0.50 is used only for diagnostic error analysis. At this threshold, the selected model produced 5,880 true negatives, 5,282 false positives, 1,705 false negatives, and 3,096 true positives. Threshold accuracy was 0.562, below the majority-class accuracy of approximately 0.699. The model is therefore not presented as a reliable automatic binary decision engine.

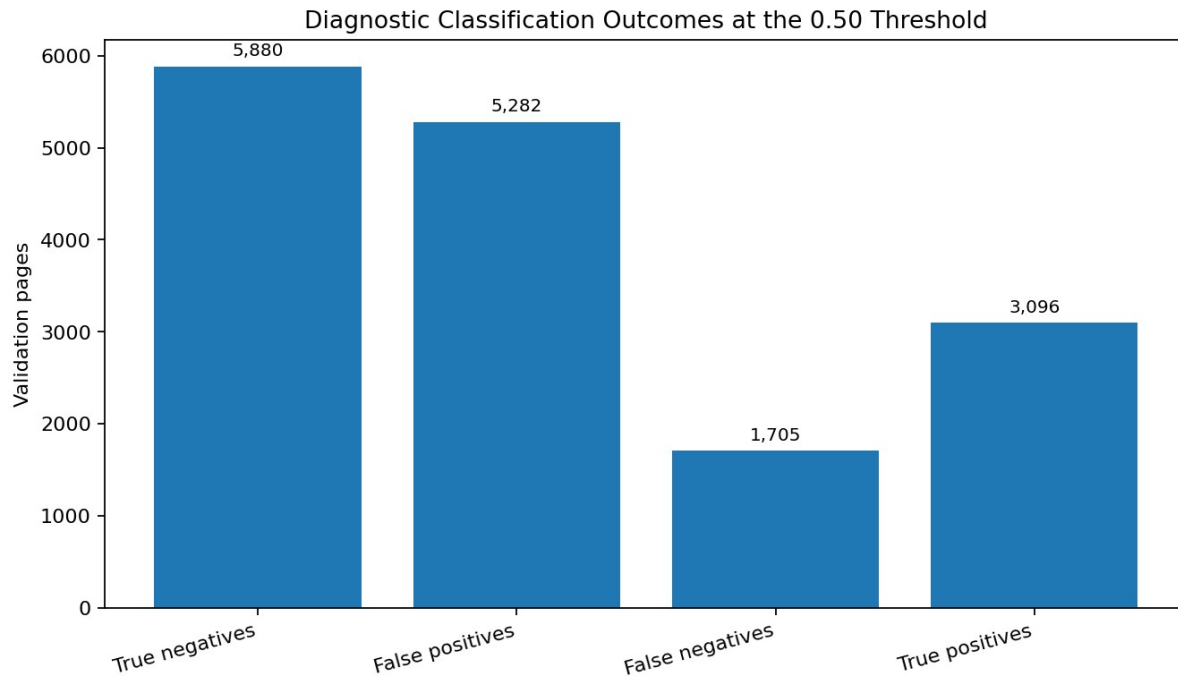


Figure 4. Diagnostic classification outcomes at the 0.50 threshold. Note. The operational product is a ranked queue evaluated by Precision@50, not a threshold-based automatic action.

Table 7. Error interpretation

Error type	Observed definition	Possible explanation	Operational response
False positive	Model score at or above 0.50, but later proxy is negative	Weak CTR or volatility may be temporary, query-specific, technical, or offset by other demand	Inspect queries, SERP features, tracking, technical status, and business role before editing
False negative	Model score below 0.50, but later proxy is positive	Decline can occur without extreme pre-outcome signals; demand and competitors are unobserved	Use scheduled portfolio monitoring in addition to the ranked top queue
Top-50 negative	High rank, but later proxy is negative	The rank identifies review priority, not certainty	Do not publish changes automatically; retain a human gate

The strongest false positives observed in the audit often combined zero feature-window clicks, Page-1 positions, and high position volatility. Some false negatives were close to the 0.50 threshold and had reasonable CTR or strong positions. These examples show that the model leans on weak engagement relative to visibility, but page-level context remains necessary.

7. Discussion

7.1 What the Model Adds Beyond the Baseline

The baseline encodes one clear editorial intuition: pages with weaker-than-expected CTR and meaningful visibility deserve review. Logistic Regression measured a stronger top-50 queue by combining the same general signal family with clicks, average position, active history, volatility, and a categorical position context. The result suggests that a learned linear ranking can refine a sensible rule without requiring a less transparent ensemble.

7.2 Why Grouped Validation Matters

Pages belonging to one client can share site architecture, tracking patterns, content templates, market conditions, and editorial practices. Random row splitting can therefore allow related pages into both training and validation. The grouped holdout directly tests whether patterns learned from 24 clients transfer to eight unseen client groups. The grouped Precision@50 happened to exceed the random-row result in this run; honesty comes from the question the split answers, not from expecting a lower score.

7.3 Relationship to Prior Research

The model's reliance on CTR and position is consistent with the broader observation that visibility and engagement should be interpreted together [1], while Manetti's study supports the broader use of observable page and ranking signals but also cautions that associations are small and non-causal [2]. The result aligns methodologically with learned reranking work that compares a model with an existing candidate ordering [3]. Organisational SEO research also supports systematic content and keyword strategy, suggesting that the queue is useful only when embedded in a repeatable review process [4].

Emerging AI-search research further suggests that future visibility systems will need to monitor structured evidence, semantic clarity, source trust, and answer-level exposure [5], [6], [9]. The current model does not measure citation in generative answers and should not be described as a GEO model. Its validated scope is organic-search content-review prioritisation.

7.4 Practical Significance

For a fixed review capacity of 50 pages, the measured increase from 0.520 to 0.640 corresponds to approximately six additional pages meeting the later decline proxy in the selected queue. This is an operational planning benefit, not a monetary ROI estimate. The value depends on whether human reviewers can correctly diagnose the page and choose an appropriate action.

8. Ranked Content Action Playbook

The model score should be translated into a transparent review plan rather than a single generic instruction to refresh. The action mapping below combines measured risk with training-referenced visibility, CTR, position, volatility, and evidence quality.

Table 8. *Archetype-to-action mapping*

Action	Reason code	Trigger	Required human review
CTR Review	CTR_GAP_ HIGH_VISIBILITY	High measured risk, Page-1 visibility, strong impressions, CTR below training band reference	Inspect titles, descriptions, query intent, SERP features, and snippet-message match
Expand and Optimise	HIGH_VALUE_ PAGE_2	High measured risk with strong Page-2 visibility	Review intent coverage, missing subtopics, internal links, and competitive evidence
Refresh Review	HIGH_RISK_ HIGH_VOLATILITY	High measured risk and high training-referenced position volatility	Check technical instability, seasonality, query mix, content freshness, and competitor changes
Refresh and Protect	HIGH_RISK_ STABLE_PAGE_1	High measured risk with stable Page-1 visibility	Update carefully, preserve successful sections, and define rollback monitoring
Protect	LOW_RISK_ HIGH_VISIBILITY	Low measured risk and strong visibility	Avoid unnecessary changes; monitor and document why the page works

Action	Reason code	Trigger	Required human review
Monitor	BORDERLINE_MEASURED_RISK	Borderline risk or limited feature history	Collect more data and inspect only if business or technical context raises priority
Manual Review	HIGH_RISK_NO_CLEAR_ARCHETYPE	High measured risk without a clear pattern	Perform a full page, query, technical, and business investigation

8.1 Mandatory Human Review

1. Confirm current search intent and the visible SERP format.
2. Inspect relevant query-level evidence rather than page aggregates alone.
3. Rule out indexing, canonical, redirect, rendering, and tracking problems.
4. Verify whether the content is stale, inaccurate, incomplete, or already fit for purpose.
5. Interpret CTR in the context of position, volume, brand demand, and SERP features.
6. Check conversion, legal, compliance, brand, and business-role constraints.
7. Assess cannibalisation and internal-link relationships.
8. Estimate the risk of damaging existing visibility.
9. Define an owner, a post-change measurement window, and a rollback plan.

8.2 What Must Not Be Automated

- Publishing, rewriting, deletion, redirect, canonical, or no-index decisions.
- Title, description, heading, or structured-data changes without page-level review.
- Causal claims that refreshing a page will recover traffic.
- Exposure of client names, raw URLs, domains, private queries, or credentials.
- Overrides of editorial, legal, safety, brand, or compliance review.
- Actions when data availability or eligibility checks fail.
- Treatment of a model probability as a final editorial decision.

8.3 Monitoring and Revalidation

Table 9. Monitoring and retraining triggers

Trigger	Response
Grouped Precision@50 falls by more than 0.10	Pause broad use and repeat grouped validation
Precision@50 approaches the base rate	Stop claiming useful prioritisation
Eligible population changes by more than 30%	Audit filters, permissions, availability, and client mix
Feature missingness rises by more than five percentage points	Investigate data collection and preprocessing
Mean model score changes by more than 0.10	Audit drift and calibration
Action mix shifts by more than 20 percentage points	Review input drift and action thresholds
A new client, market, language, or content type is introduced	Validate before operational use
The full validation audit is older than 90 days	Schedule grouped revalidation

9. Limitations and Honest Framing

Table 10. Main limitations and public-safe interpretation

Limitation	Implication
Proxy target	The outcome is a later impression-decline proxy, not a direct measure of content quality, satisfaction, revenue, or causal decay.
One development month	The analysis uses March 2026 and one grouped holdout; future months may have different demand, SERP, and client distributions.
Incomplete context	The model does not observe query text, SERP composition, competitors, seasonality, factual accuracy, conversions, or business constraints.
No intervention	No randomised or quasi-experimental refresh treatment was tested, so the paper cannot claim that a recommended action causes recovery.
Related features	Position band is derived from average position; permutation importance values are not independent causal contributions.
Threshold weakness	Accuracy at 0.50 is below the majority-class benchmark, so the model should not be used as an automatic classifier.
Operational cost	Review time and visibility are planning proxies; the study does not estimate monetary return.

The strongest defensible claim is therefore: On one grouped-client holdout, Logistic Regression measured Precision@50 of 0.640 compared with 0.520 for the frozen ML-07 baseline, an absolute measured difference of 0.120. This observed result supports directional human decision-support for content-review prioritisation within the March 2026 evaluation design. It does not establish causal refresh impact, guarantee future performance, or support automatic content changes.

10. Future Work

- Use rolling time-based and repeated grouped-client validation across several months.
- Evaluate calibration and top-K stability under changes in client mix and search demand.
- Run a prospective refresh experiment with matched controls or randomised treatment where operationally feasible.
- Add query-level and SERP-feature context under a public-safe data contract.
- Measure downstream editorial cost, conversion value, and risk of damaging high-performing pages.
- Test the action playbook separately by content type, market, and language.
- Develop separate AI-search visibility measures for citation, mention persistence, and source selection rather than treating organic ranking as a GEO outcome.

11. Conclusion

This study demonstrates a complete and auditable workflow for refresh and content opportunity scoring. It begins with a page-level data contract, separates prediction and outcome windows, preserves a transparent rule baseline, evaluates learned models on unseen client groups, demonstrates leakage, inspects errors, and translates the selected score into human-reviewed actions.

Logistic Regression measured the strongest top-50 prioritisation in the retained grouped-client validation, with Precision@50 of 0.640 versus 0.520 for the frozen baseline. Random Forest did not earn its additional complexity. The selected model relied most on CTR, position context, and volatility, but these signals remain associations rather than causes.

The practical conclusion is deliberately narrow: historical search signals can support a better first-pass review queue within the measured March 2026 design. The final decision must remain with a human who can inspect queries, technical conditions, content quality, business value, and change risk.

12. Reproducibility and Data Availability

The public repository contains the executed weekly notebooks, capstone notebook, aggregate metric receipts, paper figures, static research page, and regeneration instructions. Dataset access requires an authorised read token stored securely outside the notebook. Raw data are not included in the repository.

Repository: <https://github.com/Zafar488/flyrank-ml-internship>

Live paper: <https://zafar488.github.io/flyrank-ml-internship/>

```
git clone https://github.com/Zafar488/flyrank-ml-internship
cd flyrank-ml-internship
pip install -r requirements.txt
jupyter nbconvert --execute --to notebook --inplace work/notebooks/capstone.ipynb
```

Reproducibility constants are random seed 42, grouped validation fraction 0.25, feature window 1-15 March 2026, and outcome window 16-31 March 2026.

12.1 Software Environment

The release was executed in Google Colab with Python 3.12. The repository requirements file declares the minimum dependency versions shown below. Exact runtime versions should also be captured with `pip freeze` when a DOI release is deposited.

Table 11. *Software environment and declared dependency floor*

Component	Release setting
Execution environment	Google Colab
Python	3.12
Random seed	42
Validation	GroupShuffleSplit by anonymised client
Validation fraction	0.25
pandas	>= 2.2
NumPy	>= 1.26
scikit-learn	>= 1.4
matplotlib	>= 3.8
DuckDB	>= 1.0
huggingface_hub	>= 0.24

Repository requirements: <https://github.com/Zafar488/flyrank-ml-internship/blob/main/requirements.txt>

13. Acknowledgments and Data Credit

Built on the [FlyRank ML Internship dataset](#). The author thanks the FlyRank internship team for the anonymised research release, weekly notebook structure, and emphasis on honest validation, public-safe reporting, and human-reviewed recommendations.

14. Declarations

Declaration	Statement
Conflict of interest	The author declares no conflict of interest.
Funding	No external funding was received for this study.
Ethics	The analysis uses anonymised, non-personal search-performance data.
Author contribution	Zafar Ullah conducted the analysis, validation, interpretation, and manuscript preparation.
License	Creative Commons Attribution 4.0 International (CC BY 4.0)
AI assistance disclosure	Generative AI tools were used to support code drafting, document structuring, language refinement, and quality checks. The author reviewed the analysis, verified the reported results against executed notebooks, and accepts responsibility for the final manuscript.
Publication status	Pre-DOI technical report. The DOI field will be updated after a repository DOI is reserved and the final record is published.

References

- [1] FlyRank, The State of AI-Driven SEO in Numbers, data report, March 2026. [Research paper](#)
- [2] G. Manetti, "Google Italy Ranking Factors 2026: A Correlational Analysis of 6,824 SERP Pages Across 57 Technical and On-Page Signals," PerseoDesign Research, preprint, June 2026. [doi:10.5281/zenodo.20797976](https://doi.org/10.5281/zenodo.20797976)
- [3] A. Alhenshiri, A. Jelwal, and H. Badesh, "Behavior-Driven Semantic Re-Ranking for Personalized Web Search," Journal of Technology Research, vol. 4, no. 1, pp. 1-11, 2026. [doi:10.26629/jtr.2026.01](https://doi.org/10.26629/jtr.2026.01)
- [4] M. Reji, "Success Factors in Search Engine Optimization: A Quantitative Study of Technology Firm's Adaptability to Algorithm Changes," MSc dissertation, Dublin Business School, January 2026. [Institutional repository](#)
- [5] Y. Wen, N. Zhang, H. Yuan, X. Chen, H. Zhang, and H. Guo, "Position: Generative Engine Optimization Creates Underexamined Risks, Governance Must Target Concentration, Disclosure, and Academic Blind Spots," Proceedings of the 43rd International Conference on Machine Learning, PMLR 306, 2026. [arXiv:2606.12439](https://arxiv.org/abs/2606.12439)
- [6] S. Muñoz Tebar, "Large Language Models and the Future of SEO: Optimization Strategies for AI-Driven Search Ecosystems," technical report, 2026. [doi:10.13140/RG.2.2.15044.49289](https://doi.org/10.13140/RG.2.2.15044.49289)
- [7] P. D. Zulkarnain, Y. Hermawan, B. H. Rainanto, I. Permatasari, A. K. Fauziah, J. Van Melle, A. T. Bon, and V. Meledge-Ade, "On-Page Techniques for Improved Google Search Engine Ranking: A Tourism Village Approach," International Conference on Sustainable Tourism and Entrepreneurship, vol. 1, no. 1, pp. 7-19, 2026. [SINTA conference record](#)
- [8] Akash Raj M, Ashwin V, and Anitha Moses V, "SmartSEO: An Intelligent Machine Learning Powered System for Automated Search Engine Optimization," International Journal of Applied Smart Interdisciplinary Technologies, vol. 1, no. 1, pp. 8-13, Mar. 2026. [Article page](#)
- [9] D. Kargaev, "The SEO-to-GEO Gap: Quantifying Ranking Factor Divergence Between Traditional and Generative Search," SSRN working paper, 2026. [SSRN 6476021](https://ssrn.com/abstract=6476021)
- [10] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," Journal of Machine Learning Research, vol. 12, pp. 2825-2830, 2011. [JMLR](https://jmlr.org/)
- [11] L. Breiman, "Random Forests," Machine Learning, vol. 45, pp. 5-32, 2001. [doi:10.1023/A:1010933404324](https://doi.org/10.1023/A:1010933404324)
- [12] J. Davis and M. Goadrich, "The Relationship Between Precision-Recall and ROC Curves," in Proceedings of the 23rd International Conference on Machine Learning, pp. 233-240, 2006. [doi:10.1145/1143844.1143874](https://doi.org/10.1145/1143844.1143874)

Appendix A. Final Submission Checklist

- The title and author name appear on the first page.
- The abstract contains the question, method, measured result, intended use, and limitation.
- The baseline and models use the same grouped validation rows.
- The validation base rate is reported beside Precision@50.
- No client names, raw URLs, private queries, credentials, or pseudonymous IDs appear.
- All claims use observed, measured, directional, and decision-support language.
- All figures have captions and remain readable in print.
- The PDF contains searchable text.
- The repository and live page have public URLs.
- The AI assistance disclosure is present and assigns final responsibility to the author.
- The software environment and dependency floor are documented.

The reserved DOI should be added to the title page, suggested citation, PDF metadata, live HTML metadata, and final repository record before Version 1.2 is published.